# Keshav Sharma

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## EDUCATION

**Vellore Institute of Technology** — **Chennai, Tamil Nadu**

*Bachelor of Technology in Computer Science (Data Science Specialisation)* — *Graduation: May 2027*

## SKILL SUMMARY

**Languages:** Java, Python, C/C++, SQL, JavaScript, R

**Software Engineering:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Backend Development, REST APIs, Design Patterns, Multithreading & Concurrency, Unit Testing, Test-Driven Development (TDD), Debugging

**Generative AI & Search:** Large Language Models (LLMs), Generative AI, Retrieval-Augmented Generation (RAG), Prompt Engineering, AI Agents, Tool/Function Calling, Embeddings, Vector Search, Semantic Search, Hybrid Search, Information Retrieval, NLP, Text Processing

**Backend & Frameworks:** Spring Boot, Flask, FastAPI, Node.js, React.js, LangChain, LlamaIndex

**Databases & Search:** MongoDB, MySQL, SQLite, Redis, Vector Databases, Elasticsearch

**Cloud & DevOps:** AWS, Docker, Git, GitHub, GitHub Actions

**Tools:** VS Code, IntelliJ IDEA, Jupyter Notebook

## EXPERIENCE

***VOIS for Tech University Engagement Program*** **|LINK** — ***Aug – Sep, 2025***

- Engineered a modular data processing pipeline to preprocess, analyze, and query two real-world datasets containing 8,500+ records using LLM-powered natural language interfaces.
- Designed and implemented automated validation and response extraction workflows, improving the consistency and reliability of generated outputs across diverse user queries.
- Optimized data processing by integrating preprocessing, prompt generation, response parsing, and result validation into a reusable workflow, reducing manual analysis effort and improving scalability.

## PROJECTS

***RedisLite*** **|LINK**

- Engineered a Redis-inspired in-memory key-value database in Python with a custom RESP parser, multithreaded TCP server, thread-safe storage, O(1) LRU caching, and TTL expiration for concurrent request processing.
- Implemented Write-Ahead Logging (WAL), atomic log compaction, and automatic crash recovery to ensure durable persistence and data consistency under failures and concurrent workloads.
- Developed 46 PyTest tests covering concurrency, edge cases, and recovery scenarios; benchmarked 619K SET ops/sec in-memory and 2.8K ops/sec with WAL persistence to validate reliability and performance.

***API Rate Limiter Library*** **|LINK**

- Built a production-ready Python rate-limiting library implementing five throttling algorithms, Redis integration, and reusable Flask/FastAPI middleware for reliable API traffic control.
- Designed a modular architecture with pluggable storage backends and thread-safe concurrent request handling, enabling flexible deployment across application environments.
- Developed 30+ PyTest tests covering concurrency, edge cases, and algorithm correctness, and packaged the library on PyPI for public installation and reuse.

***AI-Powered Expense Receipt Scanner & Analytics Platform*** **|LINK**

- Built a full-stack expense management platform using FastAPI, React, SQLAlchemy, and SQLite, integrating EasyOCR, OpenCV, and NLP-based parsing to automate receipt processing and reduce processing time by 40%.
- Engineered secure backend APIs with JWT authentication, Google OAuth, CRUD operations, analytics, budget tracking, and data export, and deployed the application on Render and Vercel for reliable end-to-end operation.

***Docwise*** **|LINK**

- Engineered Docwise, an AI-powered PDF Question Answering system using Google Gemini and Retrieval-Augmented Generation (RAG), achieving 95%+ query accuracy and reducing manual document search time by 40%.
- Built a scalable PDF processing pipeline with intelligent chunking, TF-IDF vectorization, real-time indexing, prompt engineering, and hallucination-aware response validation, supporting 200+ page documents with session-based chat history.

## Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional **|LINK** — **2026**
- AWS APAC Solutions Architecture virtual experience program | **LINK** — **2025**
- Accenture Software Engineering Job Simulation | **LINK** — **2025**